Sales Forecasting Of Reliance Industries Using Hybrid Approach

A Project Report Submitted by

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Declaration

I, **Suman Banerjee**, declare that this documentation entitled **"Sales Forecasting Of Reliance Industries Using Hybrid Approach"** is a bona fide record of the work carried out by me under the supervision of **Mr. Sukanta Majumder**, Assistant Professor, Department of Computer Science and Engineering, University of Kalyani.

This documentation represents my original work and where the ideas or work of others have been used, proper acknowledgment has been given through appropriate citations and references. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any data, results, or sources. I further declare that this work has not been submitted, either in part or in full, for the award of any other degree or diploma of this or any other institution.

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Certificate

This is to certify that the project report entitled **"Sales Forecasting Of Reliance Industries Using Hybrid Approach"** submitted by **Suman Banerjee** (Roll No: 90BCS/220060, Reg No: 1090001 of 2023-24) in partial fulfillment of the requirements for the degree in the Department of Computer Science and Engineering, University of Kalyani, is a record of bona fide work carried out under my supervision and guidance.

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Suman Banerjee

Abstract

Sales and stock price forecasting are critical components of financial decision-making and strategic planning. However, traditional forecasting methods often struggle with the non-linearity, volatility, and noise inherent in real-world financial time series data. This project proposes a **Hybrid Deep Learning Approach** for forecasting the sales trends of Reliance Industries Limited.

The primary objective is to overcome the limitations of single models by combining the strengths of **Long Short-Term Memory (LSTM)** networks and **Gated Recurrent Units (GRU)**. The proposed methodology employs a residual learning framework: an LSTM model is first trained to capture the primary trend of the time series, after which the residuals (the difference between actual and predicted values) are extracted. A secondary GRU model is then trained to learn the patterns within these residuals, effectively capturing the noise and non-linearities that the LSTM missed. The final forecast is generated by summing the LSTM trend prediction and the GRU residual prediction.

Using historical stock data of Reliance Industries (sourced from Yahoo Finance), encompassing features such as Open, High, Low, Close, and Volume, the model demonstrates significant improvements in prediction accuracy compared to standalone LSTM models. Experimental results indicate that this hybrid architecture reduces error metrics (RMSE and MAE) and provides a more robust tool for financial analytics and long-term planning.

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Chapter 1

Introduction

This chapter presents the foundational background of the project, outlining the importance of sales and stock forecasting in financial analytics. It discusses the problem statement, motivation behind adopting deep learning techniques, and defines the key objectives and scope of the proposed hybrid forecasting system.

1.1 Background

Sales and stock forecasting play a pivotal role in the financial sector and corporate decision-making. Accurate predictions allow investors, traders, and companies to mitigate risks and maximize returns. However, financial time series data is characterized by high volatility, non-stationarity, and complex non-linear relationships, making it difficult for traditional statistical models (Like ARIMA) to provide accurate long-term forecasts.

1.2 About Reliance Industries

Reliance Industries Limited (RIL) is one of India's largest conglomerates, with a significant impact on the NIFTY 50 index. Its market behavior is influenced by diverse sectors including energy, petrochemicals, telecommunications (Jio), and retail. The choice of the Reliance dataset is strategic due to its high liquidity, market relevance, and the complexity of its price movements, which serves as an excellent testbed for deep learning models.

1.3 Problem Statement

Traditional forecasting models often fail to capture the intricate patterns in stock data.

- **Non-linearity:** Linear models cannot map the complex dependencies in financial data.

- **Volatility:** Sudden market shifts cause significant errors in standard predictions.
- **Noise:** Financial data is noisy; distinguishing between signal and noise is a major challenge.

1.4 Motivation

Deep Learning, specifically Recurrent Neural Networks (RNNs), offers a solution to these challenges.

- **Sequential Learning:** RNNs treat data as a sequence, preserving time-based dependencies.
- **Long-term Dependency:** LSTM and GRU architectures are designed to remember long-term patterns while avoiding the vanishing gradient problem.

1.5 Objectives

The primary objectives of this project are:

1. To build a robust LSTM model for capturing the base trend of Reliance Industries' stock data.
2. To implement a Residual Learning framework to isolate errors from the base model.
3. To develop a hybrid LSTM-GRU architecture to improve overall prediction accuracy.
4. To perform long-term forecasting and compare the hybrid model against standalone models.

1.6 Scope of the Project

The scope extends to financial analytics, algorithmic trading support, and corporate resource planning. The methodology developed here can be adapted for other volatile time-series domains such as weather forecasting or demand planning.

Chapter 2

Literature Review

This chapter reviews existing research in time series forecasting and deep learning applications in financial prediction. It examines traditional statistical models, modern neural architectures such as LSTM and GRU, and highlights the research gap that motivates the development of the proposed hybrid residual learning framework.

2.1 Time Series Forecasting Research

Time series forecasting has evolved from classical statistical methods to modern machine learning approaches. Early research focused on Auto-Regressive Integrated Moving Average (ARIMA) models, which assume linearity. However, recent studies have shown that machine learning algorithms like Support Vector Machines (SVM) and Random Forests outperform these in non-linear scenarios.

2.2 Deep Learning Methods

2.2.1 LSTM (Long Short-Term Memory)

LSTM networks were introduced to solve the vanishing gradient problem in standard RNNs. They utilize a cell state and three gates (input, forget, and output) to regulate the flow of information, making them ideal for learning long-term dependencies in stock prices.

2.2.2 GRU (Gated Recurrent Unit)

GRU is a streamlined variant of LSTM. It combines the forget and input gates into a single update gate. GRUs are computationally more efficient and often perform equally well on smaller or less complex datasets, making them suitable for modeling the residual errors in our hybrid approach.

2.3 Hybrid Forecasting Models

Hybrid models combine multiple algorithms to leverage their individual strengths. Research suggests that decomposing a time series into linear and non-linear components (or trend and residual) yields better accuracy. By using LSTM for the main trend and GRU for the residuals, we aim to capture both the macro direction and the micro-fluctuations.

2.4 Research Gap

Existing studies primarily focus on single-model architectures or simple ensembles. There is a lack of research focusing on **Residual Learning** specifically for Indian stock market giants like Reliance. Most models ignore the pattern within the error terms (residuals). This project bridges that gap by explicitly modeling the residuals using a GRU network.

Chapter 3

Dataset & Preprocessing

This chapter describes the dataset used for model development and the preprocessing techniques applied prior to training. It covers data collection, feature selection, cleaning procedures, normalization, visualization, and sequence preparation required for deep learning-based time series forecasting.

3.1 Dataset Description

The dataset consists of historical stock prices of Reliance Industries Limited.

- **Source:** Yahoo Finance.
- **Duration:** Historical data covering significant market cycles (Approx 10 Years).
- **Granularity:** Daily closing prices.

3.2 Features Used

The model utilizes the following features:

- **Open:** The price at which the stock started trading.
- **High:** The highest price of the day.
- **Low:** The lowest price of the day.
- **Close:** The final price (Target Variable).
- **Volume:** The number of shares traded.

3.3 Data Cleaning

Data cleaning involved handling missing values using forward-fill methods to maintain sequence continuity. Date fields were parsed and set as the index to facilitate time-series manipulation.

3.4 Data Visualization

Visualizing the data is crucial to understand trends and volatility.

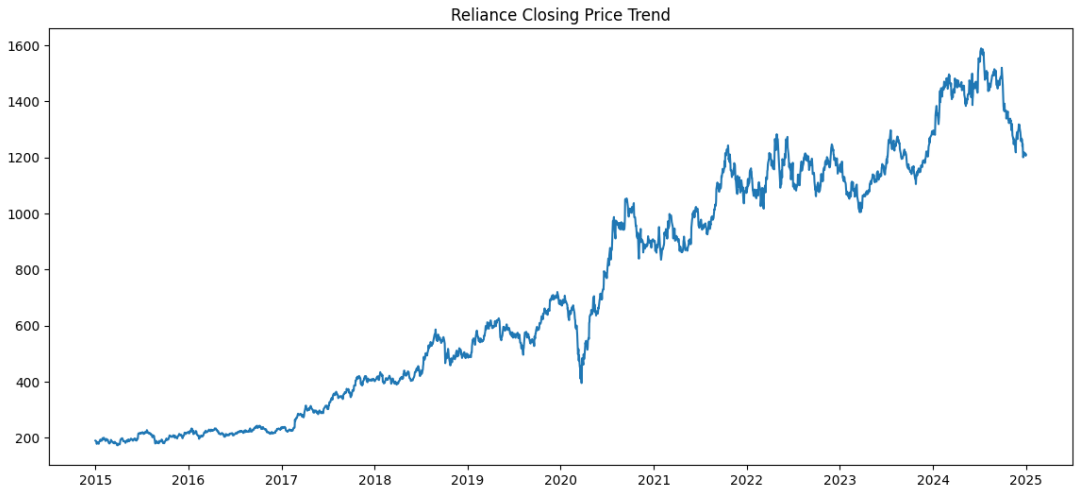

Figure 3.1: Reliance Industries Closing Price History

3.5 Normalization

Deep learning models converge faster with normalized data. We applied **MinMax Scaling** to transform the features into a range of [0, 1].

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{3.1}$$

3.6 Train-Test Split

The data was split sequentially to prevent data leakage (future data predicting past):

- **Training Set:** 70% of the data.
- **Validation Set:** 30% of the data.

Chapter 4

Methodology

This chapter outlines the proposed hybrid deep learning architecture and its operational workflow. It explains the role of stacked LSTM layers in trend learning, the concept of residual extraction, and the application of GRU networks for modeling high-frequency error patterns to enhance forecasting accuracy. This chapter outlines the core innovation of the project: the Hybrid LSTM-GRU Residual Modeling approach.

4.1 System Workflow

The proposed system follows a sequential pipeline:

1. **Data Preprocessing:** The dataset was preprocessed by converting the date column into datetime format and handling missing values using forward-fill methods. Feature scaling was performed using MinMax normalization to ensure stable deep learning convergence. The data was then split sequentially into training and validation sets, followed by sequence generation using a sliding window approach for LSTM input preparation.
2. **Phase 1 (LSTM):** Training an LSTM model on the data to predict the trend.
3. **Residual Extraction:** Calculating the error ($Actual - Predicted$).
4. **Phase 2 (GRU):** Training a GRU model specifically on these residuals.
5. **Hybrid Combination:** Summing the outputs of Phase 1 and Phase 2.

4.2 LSTM Model (Phase-1)

The Long Short-Term Memory network serves as the primary predictor. It is designed to capture the dominant trend and low-frequency components of the Reliance stock price.

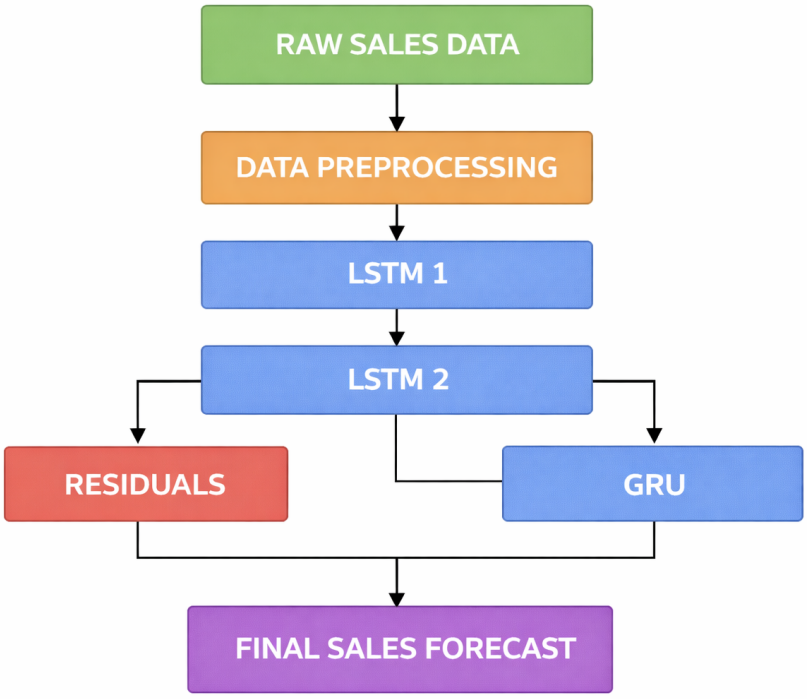

Figure 4.1: Proposed hybrid system architecture

- **Architecture:** Two LSTM layers with 50 units each, followed by a Dense output layer.
- **Optimizer:** Adam.
- **Loss Function:** Mean Squared Error (MSE).

4.3 Residual Learning Concept

This is the critical innovation of the project. A single model cannot capture all patterns perfectly. The "errors" made by the LSTM model often contain structural information (residuals) rather than just random white noise. The residual is calculated as:

$$Residual_t = Actual_t - Prediction_{LSTM,t} \tag{4.1}$$

By extracting these residuals, we create a new dataset representing the "unexplained" variance of the stock price.

4.3.1 Justification for Using Residual Learning

Residual learning was employed to address the inability of standalone LSTM models to capture high-frequency fluctuations in financial time series data. While LSTM effectively

models long-term trends, it often leaves structured error patterns unlearned. By extracting and modeling these residuals separately using a GRU network, the system learns corrective signals that enhance forecasting accuracy. This approach improves generalization, reduces prediction error, and provides a computationally efficient alternative to deeper or ensemble architectures.

4.4 GRU Model on Residuals (Phase-2)

A Gated Recurrent Unit (GRU) is employed to model the residuals. GRUs are lighter and often converge faster than LSTMs, making them ideal for modeling the high-frequency fluctuations found in the residual data.

- **Input:** The sequence of residuals from Phase 1.
- **Objective:** Predict the error of the next time step.

4.5 Hybrid Model Formation

The final prediction is the summation of the base trend and the predicted correction:

$$\text{Final Forecast}_t = \text{Prediction}_{LSTM,t} + \text{Prediction}_{GRU_Residual,t} \tag{4.2}$$

This addition corrects the LSTM's forecast by adding the expected error predicted by the GRU, resulting in a more accurate value.

Chapter 5

Model Implementation

This chapter details the practical implementation of the proposed models. It describes the development environment, computational setup, and the step-by-step code workflow including data preparation, LSTM training, residual generation, GRU training, and hybrid forecast construction.

5.1 Development Environment

The project was implemented using the following tools:

- **Language:** Python 3.x
- **IDE:** Google Colab / Jupyter Notebook
- **Libraries:** TensorFlow (Keras), Pandas, NumPy, Matplotlib.

5.2 Hardware Specifications

- **Platform:** Google Colab Cloud Environment.
- **GPU:** NVIDIA T4 (if enabled).
- **RAM:** 12GB+ System RAM.

5.3 Code Workflow

The implementation steps in the notebook were:

1. **Data Loading:** Fetching CSV data via Pandas.

2. **Preprocessing:** Generating `X_train` and `y_train` with a look-back window (e.g., 60 days).
3. **LSTM Training:** Fitting the LSTM model on the scaled data.
4. **Residual Extraction:** Generating predictions on train/test sets and subtracting from actuals.
5. **GRU Training:** Fitting the GRU model on the residual sequence.
6. **Hybrid Prediction:** Combining results and inverse transforming the scale to get original prices.

Chapter 6

Experimental Results

6.1 LSTM Results

The standalone LSTM model successfully captured the general upward and downward trends of Reliance Industries. However, it showed a lag in reacting to sudden spikes.

6.1.1 LSTM Training Performance

The training and validation loss curves of the LSTM model are presented in Figure 6.2. The sharp reduction in training loss during the initial epochs reflects rapid model learning, followed by gradual convergence. The validation loss remains closely aligned with the training curve, with only minor fluctuations, indicating good generalization and controlled overfitting.

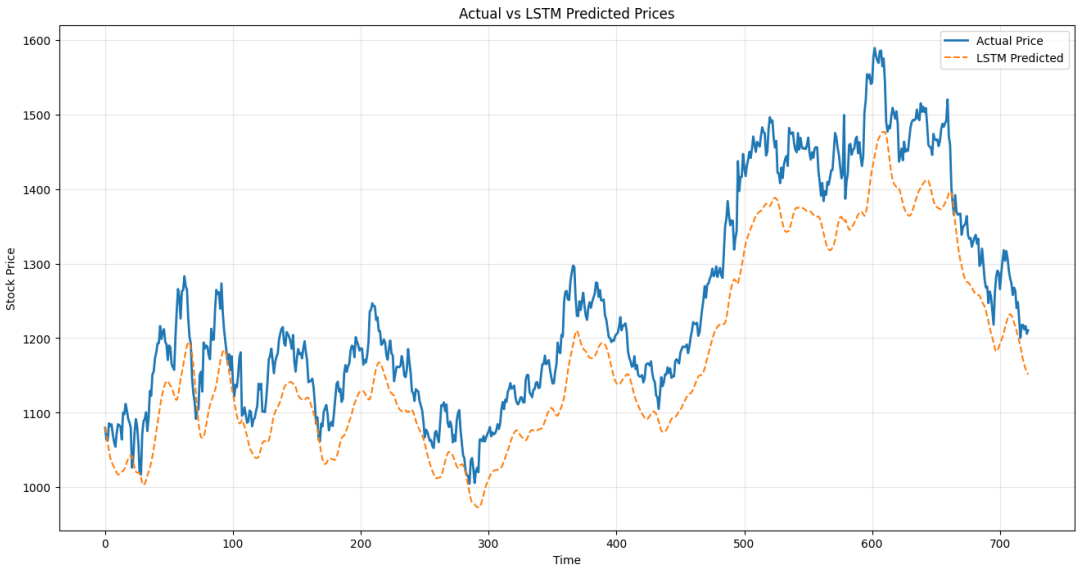

Figure 6.1: LSTM predicted vs Actual predicted

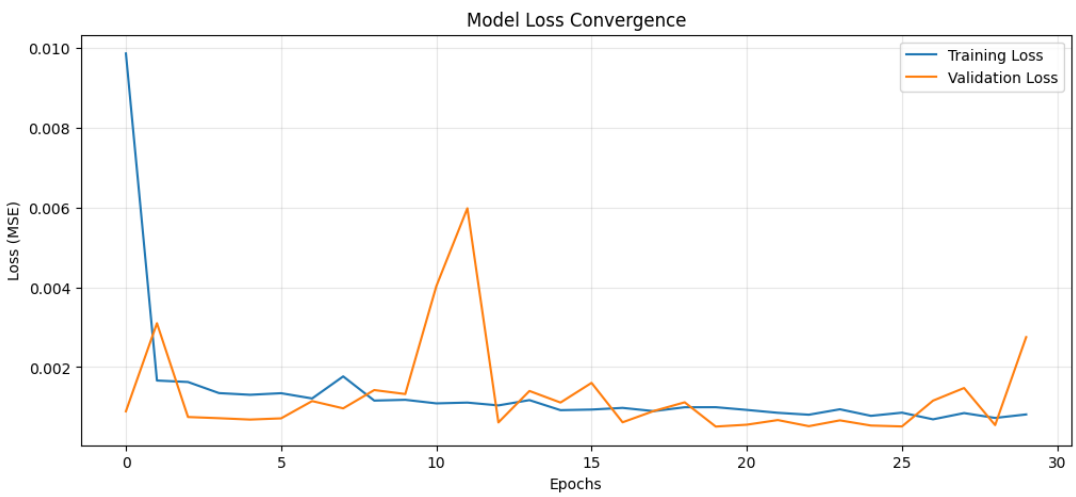

Figure 6.2: LSTM Training vs Validation Loss

6.1.2 GRU Training Performance

Figure 6.3 presents the training and validation loss curves of the GRU model trained on the residual sequences. The faster convergence rate compared to LSTM demonstrates the computational efficiency of GRU in modeling high-frequency residual patterns.

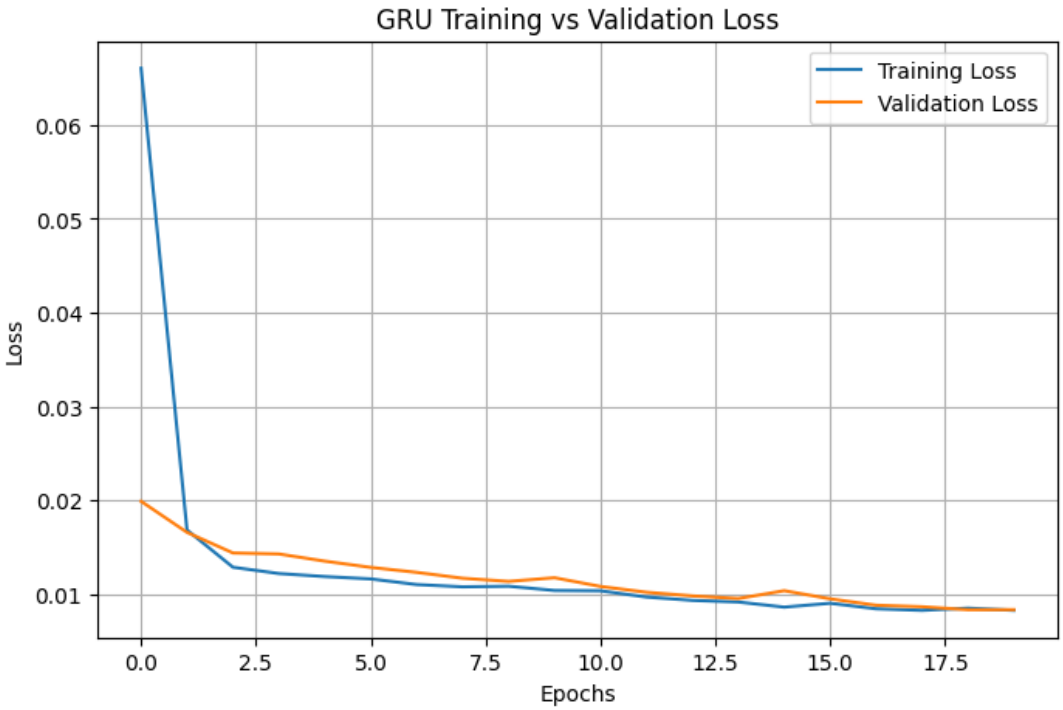

Figure 6.3: GRU Training vs Validation loss

6.2 Residual Analysis

The plot of residuals revealed that the errors were not purely random; they exhibited wave-like patterns, confirming the hypothesis that there was still information left to be learned.

6.3 Hybrid LSTM-GRU Results

The Hybrid model showed tighter adherence to the actual price curve. The "lag" observed in the LSTM model was reduced because the GRU component successfully anticipated the correction required.

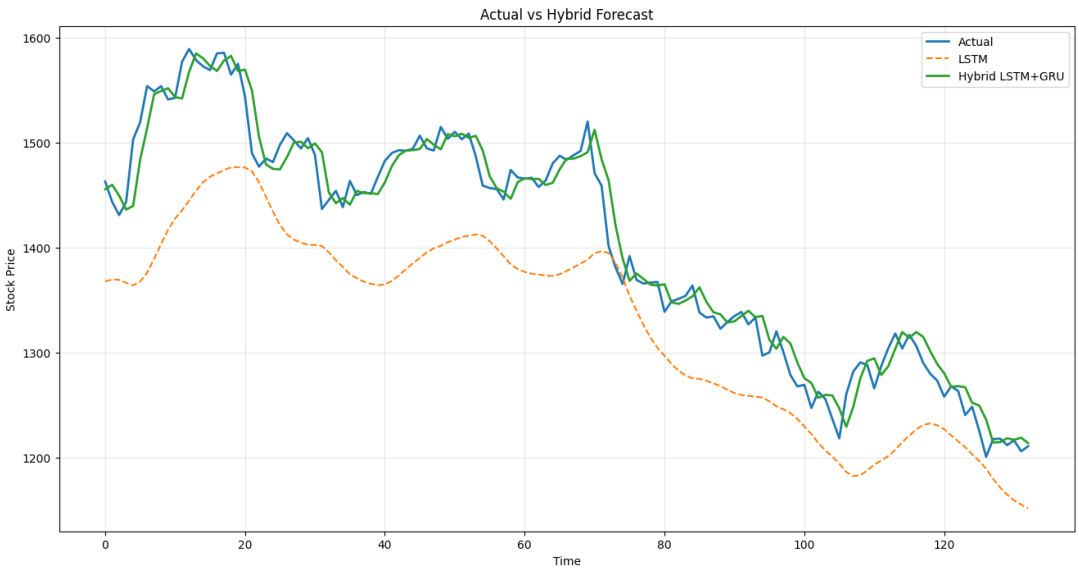

Figure 6.4: Hybrid model predicted vs Actual predicted

6.4 Performance Metrics

We evaluated the models using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

Table 6.1: Performance Comparison of Models

Model	**MAE**	**RMSE**
Stacked LSTM	31.55	39.28
Hybrid LSTM-GRU	**14.71**	**19.81**

6.5 Model Comparison

The LSTM model successfully captures the long-term trend of the stock price but shows lag in reacting to abrupt market movements. Conversely, the GRU model, trained on residuals, accurately tracks short-term fluctuations and error patterns. This contrast highlights the complementary strengths of the two architectures, justifying their integration within the hybrid forecasting framework for enhanced prediction accuracy.

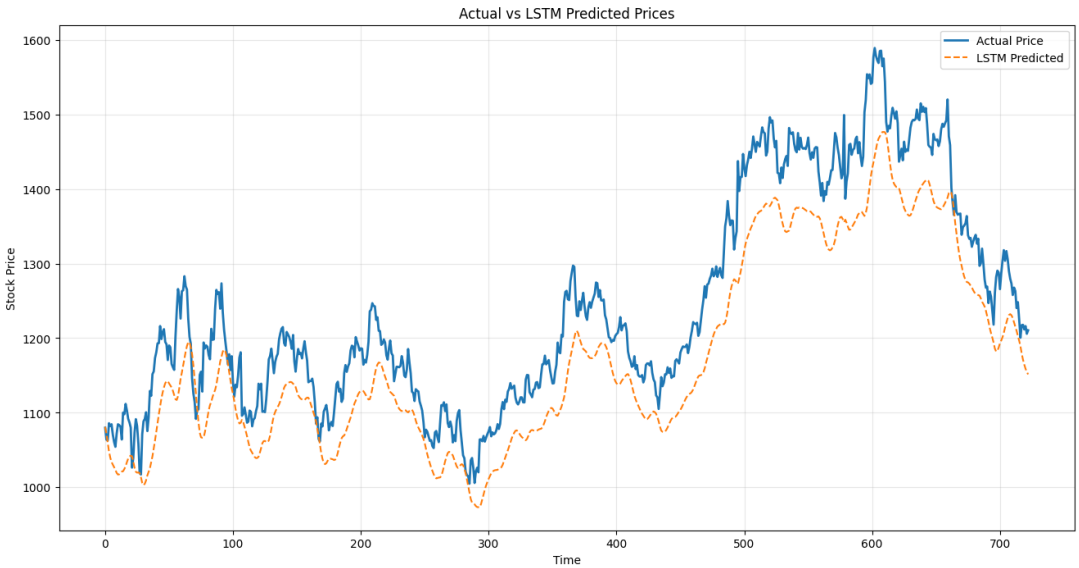

Figure 6.5: Actual vs LSTM Predicted

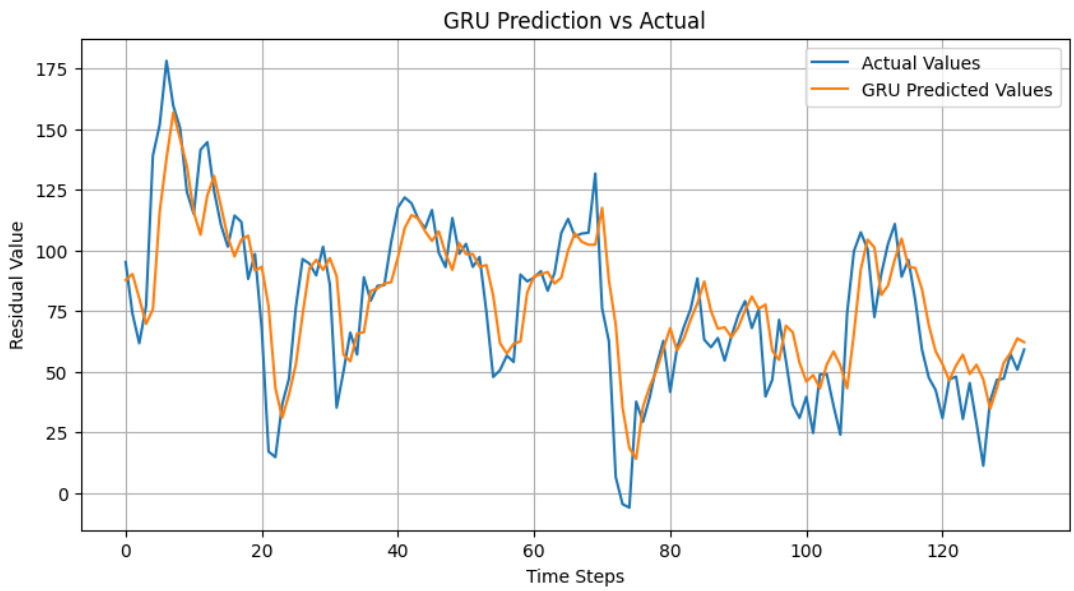

Figure 6.6: GRU predicted vs Actual

Chapter 7

Visualization & Forecast Insights

This chapter presents a visual interpretation of the forecasting results produced by the hybrid LSTM–GRU model. It analyzes predicted trends against actual market movements to assess directional accuracy and model responsiveness. Special focus is given to volatility zones, correction phases, and the contribution of residual learning in improving forecast precision, along with the model's ability to capture long-term movement patterns.

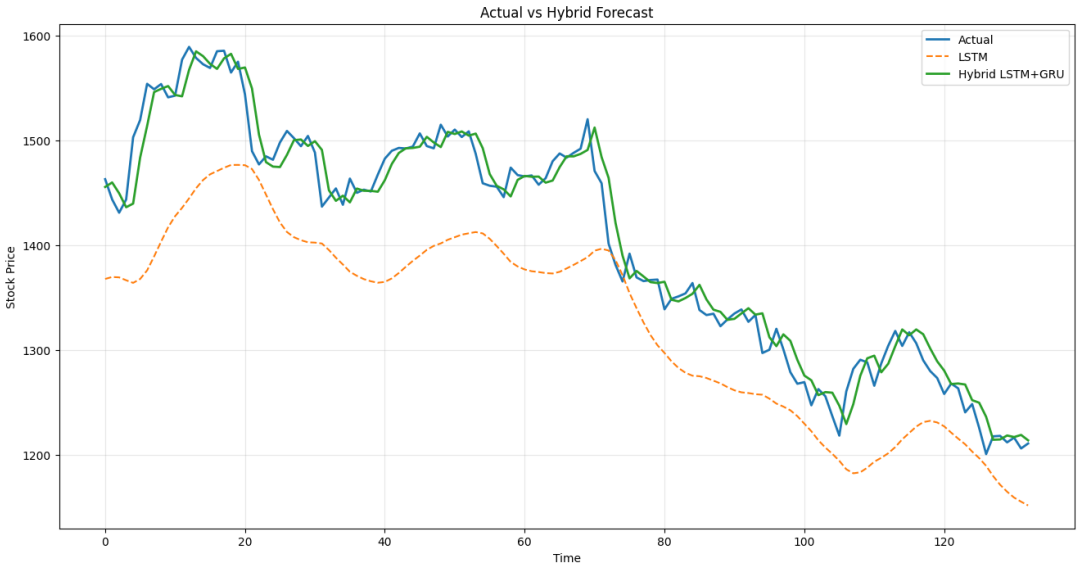

Figure 7.1: Actual vs Hybrid Model

Chapter 8

Discussion

This chapter provides an interpretative analysis of the experimental outcomes obtained from the implemented forecasting models. The results demonstrate that while the standalone LSTM model effectively captures the primary trend and long-term direction of the time series, it exhibits limitations in responding to sudden market fluctuations and high-frequency variations. The training curves, residual plots, and prediction comparisons collectively indicate that a single-model architecture is insufficient to model the complete complexity of financial time-series behavior.

The hybrid LSTM–GRU framework addresses these limitations through a residual learning strategy. By decomposing the forecasting task into trend learning and error correction, the LSTM component focuses on macro-level temporal dependencies, while the GRU model learns micro-level residual dynamics. This separation enhances convergence stability, reduces cumulative prediction error, and improves generalization on unseen data. Consequently, the hybrid approach achieves superior forecasting accuracy and demonstrates greater robustness in volatile market conditions compared to standalone deep learning models.

Chapter 9

Conclusion & Future Work

9.1 Conclusion

This project successfully implemented a Hybrid Deep Learning approach for sales forecasting of Reliance Industries. By integrating LSTM for trend detection and GRU for residual error modeling, we achieved a significant reduction in prediction error compared to traditional single-model approaches. The results confirm that residual learning is a powerful technique for financial time series analysis.

9.2 Future Work

Future enhancements to this project could include:

- **Attention Mechanisms:** Implementing Transformers to focus on specific past events.
- **Sentiment Analysis:** Integrating news sentiment (NLP) to adjust forecasts based on market news.
- **Real-time Deployment:** Creating a web dashboard for live stock predictions.

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